R13

Code No: 115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, February - 2022
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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used paper feb-22 jntu

1.a) Write a LEX program that recognizes the tokens of a C language and returns the token found.
b) Consider the following grammar and eliminate ambiguity and construct the parse tree for the sentence (a,((a,a),(a,a))). [7+8]

$S \rightarrow (L) \mid a$ $L \rightarrow L,S \mid S$

2.a) How can NFA be generated from a Regular expression? Explain the steps with the following example: (a/b)*abb.
b) Write the recursive descent parser for the following grammar: [7+8]

$E \rightarrow TE'$ $E' \rightarrow +TE' \mid \varepsilon$ $T \rightarrow FT'$ $T' \rightarrow *FT' \mid \varepsilon$

3. Compute LR(0) items for the following grammar and construct SLR parsing Table:
$S \rightarrow L=R \mid R$ $L \rightarrow *R \mid id$ $R \rightarrow L$ [15]

4. What is shift-reduce parsers? Explain briefly and mention the conflicts that occur during shift-reduce parsing. [15]

5.a) Write the Syntax Directed Definition for the following grammar:
$E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E)/id$
b) Translate the arithmetic expression a+(b+c)*d into a syntax tree, postfix notation and three address code notation. [7+8]

6.a) Construct the Syntax Directed Translation scheme to convert a given arithmetic expression into three address code.
b) Explain various symbol table organization formats and state the merits and demerits. [8+7]

7. What is data – flow analysis? Explain various terminology used in data – flow analysis and live – variable analysis. [15]

8. Explain in detail about various machine dependent code optimization techniques. [15]

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